Math 257 Definitions and Theorems

Week 5: Modules 15-18

Fall 2023

15 Subspaces

Definition 15.1. A non-empty subset H of $\mathbb{R}^n$ is a **subspace** of $\mathbb{R}^n$ if it satisfies the following two conditions:

- if $\mathbf{u}, \mathbf{v} \in H$, then the sum $\mathbf{u} + \mathbf{v} \in H$. ($H$ is *closed under vector addition*), and
- if $\mathbf{u} \in H$ and c is a scalar, then $c\mathbf{u} \in H$. (H is *closed under scalar multiplication.*)

Theorem 15.1. *Let* $\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m} \in \mathbb{R}^n$. *Then* $\mathrm{Span}\,(\mathbf{v_1}, \mathbf{v_2}, \ldots, \mathbf{v_m})$ *is a subspace of* $\mathbb{R}^n$.

16 Column Spaces and Null Spaces

Definition 16.1. The **column space**, written as $\mathrm{Col}(A)$, of an $m \times n$ matrix A is the set of all linear combinations of the columns of A. If $A = \begin{bmatrix} \mathbf{a_1} & \mathbf{a_2} & \cdots & \mathbf{a_n} \end{bmatrix}$, then $\mathrm{Col}(A) = \mathrm{Span}\,(\mathbf{a_1}, \mathbf{a_2}, \ldots, \mathbf{a_n})$.

Theorem 16.1. *Let* A *be an* $m \times n$ *matrix. Then* $\mathrm{Col}(A)$ *is a subspace of* $\mathbb{R}^m$.

Theorem 16.2. *Let* A *be an* $m \times n$ *matrix and* $\mathbf{b} \in \mathbb{R}^m$. *Then* $\mathbf{b}$ *is in* $\mathrm{Col}(A)$ *if and only if the linear system* $A\mathbf{x} = \mathbf{b}$ *has a solution.*

Definition 16.2. The **nullspace** of an $m \times n$ matrix A, written as $\mathrm{Nul}(A)$, is the set of all solutions to the homogeneous equation $A\mathbf{x} = \mathbf{0}$; that is, $\mathrm{Nul}(A) = \{\mathbf{v} \in \mathbb{R}^n \ : \ A\mathbf{v} = \mathbf{0}\}$.

Theorem 16.3. *The nullspace of an* $m \times n$ *matrix* A *is a subspace of* $\mathbb{R}^n$.

Theorem 16.4. *Let* A *be an* $m \times n$ *matrix, let* $\mathbf{b} \in \mathbb{R}^m$, *and let* $\mathbf{w} \in \mathbb{R}^n$ *such that* $A\mathbf{w} = \mathbf{b}$. *Then* $\{\mathbf{v} \in \mathbb{R}^n \ : \ A\mathbf{v} = \mathbf{b}\} = \mathbf{w} + \mathrm{Nul}(A)$.

17 Abstract Vector Spaces

Definition 17.1. A **vector space** is a non-empty set V of objects, called **vectors**, for which linear combinations make sense. More precisely, there are two operations on V, called *addition* and *multiplication* by scalars (real numbers), satisfying the following axioms for all $u, v, w \in V$ and for all scalars $c, d \in \mathbb{R}$:

- $\mathbf{u} + \mathbf{v}$ is in V. (V is "closed under addition".)
- $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$.
- $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$.
- There is a vector (called the zero vector) $\mathbf{0}_V$ in V such that $\mathbf{u} + \mathbf{0}_V = \mathbf{u}$.
- For each $\mathbf{u}$ in V, there is a vector $-\mathbf{u}$ in V satisfying $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}_V$.
- $c\mathbf{u}$ is in V. (V is "closed under scalar multiplication".)
- $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$.
- $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$.
- $(cd)\mathbf{u} = c(d\mathbf{u})$.
- $1\mathbf{u} = \mathbf{u}$.

Definition 17.2. Let V be a vector space. A non-empty subset $W \subseteq V$ is a **subspace** of V if

- $\mathbf{u} + \mathbf{v}$ for all $\mathbf{u}, \mathbf{v} \in U$ (*closed under addition*)
- $c\mathbf{u}$ for all $\mathbf{u} \in U$ and $c \in \mathbb{R}$ (*closed under scalar multiplication*)

18 Linear Independence

Definition 18.1. Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are said to be **linearly independent** if the equation

$$x_1\mathbf{v}_1 + x_2\mathbf{v}_2 + \cdots + x_p\mathbf{v}_p = \mathbf{0}$$

has only the trivial solution (namely, $x_1 = x_2 = \cdots = x_p = 0$). We say the vectors are **linearly dependent** if they are not linearly independent.

Theorem 18.1. *Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are linearly dependent if and only if there is $i \in \{1, \ldots, p\}$ such that $\mathbf{v}_i \in \mathrm{Span}(\mathbf{v}_1, \ldots, \mathbf{v}_{i-1}, \mathbf{v}_{i+1}, \ldots, \mathbf{v}_p)$.*

Theorem 18.2. *Let A be an $m \times n$ matrix. The following are equivalent:*

- *The columns of A are linearly independent.*
- *$A\mathbf{x} = \mathbf{0}$ has only the solution $\mathbf{x} = \mathbf{0}$.*
- *A has n pivots.*
- *there are no free variables for $A\mathbf{x} = \mathbf{0}$.*